Title

Longest Substring Without Repeating Characters in Java

Objective

To write a Java program to find the length and the longest substring without repeating characters in a given string.

System Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 4 GB RAM
- Keyboard and Mouse

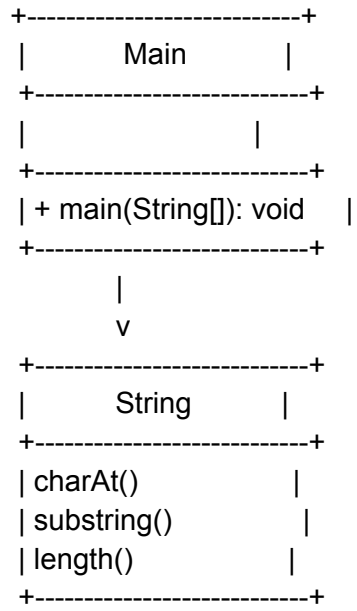
Software Requirements

- Java JDK
- VS Code / Eclipse / IntelliJ IDEA
- Windows/Linux operating system

Algorithm / Logic Description

1. Start the program.
2. Read a string from the user.
3. Initialize `maxLength`, `start`, and `bestStart` to zero.
4. Traverse each character in the string.
5. Check whether the current character already exists in the current substring.
6. If a repeating character is found, move the starting position to the next position after its previous occurrence.
7. Calculate the current substring length.
8. If the current length is greater than `maxLength`, update `maxLength` and `bestStart`.
9. Extract the longest substring using the `substring()` method.
10. Display the length and the longest substring.
11. Stop the program.

UML Class Diagram



Source Code

```
 import java.util.Scanner;


public class Main {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);


        System.out.print("Enter string: ");

        String s = sc.nextLine();


        int maxLength = 0;

        int start = 0;

        int bestStart = 0;


        for (int i = 0; i < s.length(); i++) {
```

```
            for (int j = start; j < i; j++) {
                if (s.charAt(j) == s.charAt(i)) {
                    start = j + 1;
                    break;
                }
            }

            if (i - start + 1 > maxLength) {
                maxLength = i - start + 1;
                bestStart = start;
            }
        }

        String longest = s.substring(bestStart, bestStart + maxLength);

        System.out.println("Length: " + maxLength);
        System.out.println("String: " + longest);

        sc.close();
    }
}
```

Compilation and Execution Commands

Compile:

javac Main.java

Execute:

java Main

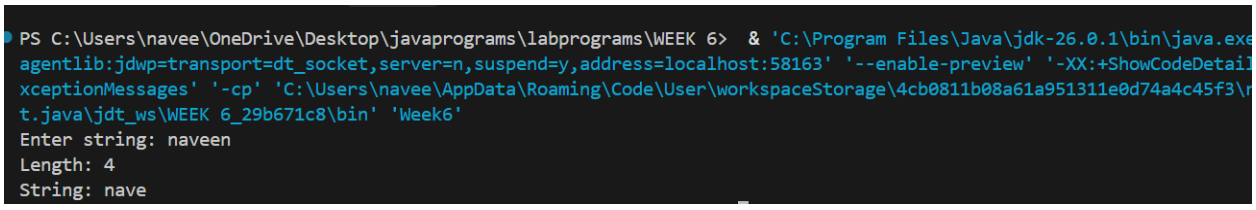
Output Screen

Example 1:

Enter string: abcabcbb
Length: 3
String: abc

Example 2:

Enter string: pwwkew
Length: 3
String: wke



Result

The Java program was successfully implemented to find the length and the longest substring without repeating characters using Java String handling methods.